System Requirements Specification: eRegistrar

Course: CS425, Software Engineering
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Project repository: `https://github.com/ElfatihZiad/eregistrar`

1. Introduction

This document specifies the requirements for **eRegistrar**, a web-based course scheduling and registration system for the Computer Science department. It covers the use-case model (actors, use-case diagram, and descriptions of the major use cases) along with the non-functional requirements.

It follows from the Vision Document (Lab 1) and feeds into the architecture (Lab 3) and use-case analysis (Labs 4 and 5).

In scope: faculty profiles, course catalog, schedule generation, and student registration. **Out of scope:** admissions, grading, tuition, and room allocation. Those stay with the university's existing student information system.

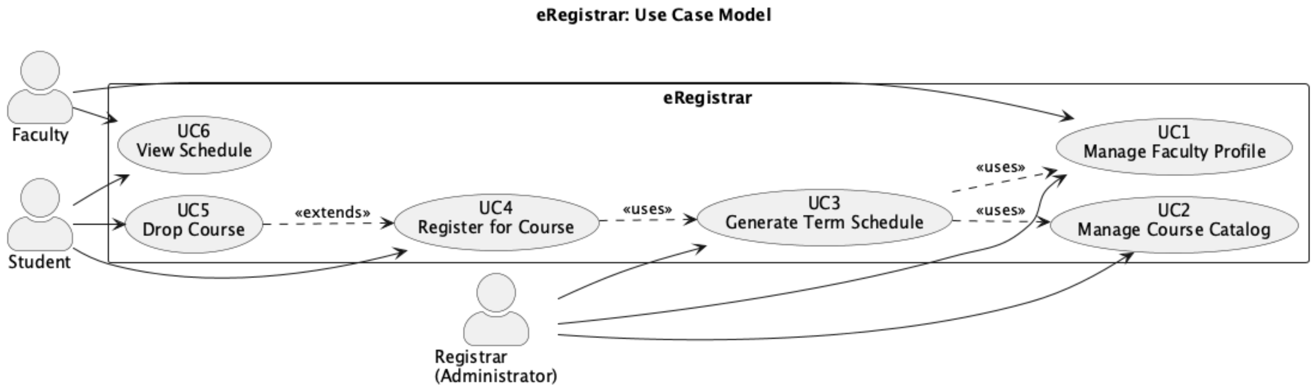
Definitions

Term	Meaning
Block	An 8-week teaching period; a course is taught within one block.
Entry	An intake cohort of students; the department runs four per year.
Section	A specific offering of a course in a block, with an assigned faculty member and a capacity.
Schedule	The set of sections offered in a term, in draft or published state.

2. Actors

Actor	Description
Student	Registers for and drops courses; views the published schedule.
Faculty	Maintains their profile (specializations, teachable courses, block availability) and views assigned sections.
Registrar (Administrator)	Maintains the catalog, generates and publishes the schedule, assigns faculty.

3. Use-Case Model



ID	Use Case	Primary Actor	Feature	Description
UC1	Manage Faculty Profile	Faculty	F1	Create, view and update a faculty profile: specializations, teachable courses, block availability.
UC2	Manage Course Catalog	Registrar	F2	CRUD for courses, prerequisites and program requirements.
UC3	Generate Term Schedule	Registrar	F3, F4	Build a draft schedule of sections for a term and assign qualified, available faculty. **Described in full below.**
UC4	Register for Course	Student	F5, F6	Register for a section of a published schedule, subject to the registration rules. **Described in full below.**
UC5	Drop Course	Student	F5	Drop a registration within the registration window, releasing the seat.
UC6	View Schedule	Student, Faculty	F7	View the published schedule: a student's own registrations, or a faculty member's assigned sections.

UC3 and UC4 are the architecturally significant use cases, and they're the ones carried into the sequence, collaboration and VOPC diagrams of Labs 4 and 5.

4. Use-Case Descriptions

UC3: Generate Term Schedule

Use Case Number	3
Name	Generate Term Schedule
Brief description	The registrar generates a draft schedule for a term. The system creates a section for each course required by the active entries and assigns a qualified, available faculty member to each, reporting any course it couldn't staff.
Actors	Registrar
Preconditions	The registrar is logged in. The term, its blocks and entries are defined, the course catalog is populated (UC2), and at least one faculty profile exists (UC1).

Basic Flow

Step	User Actions	System Actions
1	The registrar selects "Generate schedule" and chooses a term.	The system displays the generation form with the term's blocks and active entries.
2	The registrar confirms and requests generation.	The system determines the courses required by each entry in each block, placing 400-level prerequisite courses in earlier blocks and 500-level courses in later blocks.
3	N/A	For each required course the system creates a section and selects a faculty member who lists the course as teachable, is available in that block, and isn't already assigned to another section in that block.
4	N/A	The system saves the result as a **draft** schedule and displays it, along with any courses it couldn't staff.

Alternate Flows

- **A1: No qualified faculty available.** The section is created unassigned and the course is added to the unstaffed list. Generation continues.
- **A2: Schedule already exists.** The registrar has to choose explicitly between discarding the existing draft and cancelling. A published schedule is never overwritten.

Postconditions	A draft schedule exists for the term. Nothing is visible to students until it's published.
Business Rules	**BR3:** A faculty member may not be assigned to two sections in the same block. **BR4:** A faculty member may only be assigned to a course listed as teachable in their profile. **BR5:** 400-level prerequisite courses are scheduled in earlier blocks than the 500-level courses that depend on them.

UC4: Register for Course

Use Case Number	4
Name	Register for Course
Brief description	A student registers for a section of a published schedule. The system accepts the registration only if the prerequisite, capacity, time-conflict and course-load rules are all satisfied.
Actors	Student
Preconditions	The student is logged in. A schedule is published for the term and the registration window is open.

Basic Flow

Step	User Actions	System Actions
1	The student opens "Register" for the term.	The system displays the published schedule for the student's entry, with the assigned faculty and the remaining seats for each section.
2	The student selects a section and confirms.	The system validates the registration: prerequisites completed, a seat available, no time conflict in that block, and course load within the student's limit.
3	N/A	If every rule passes, the system creates the registration, reduces the available seats by one in the same transaction, and displays a confirmation with the updated schedule.

Alternate Flows

- **A1: Prerequisite not met.** The registration is refused and the missing prerequisite is named.
- **A2: Section full.** The registration is refused, including when the last seat is taken by another student between display and confirmation.
- **A3: Time conflict.** The registration is refused and the conflicting section is identified.
- **A4: Course load exceeded.** The registration is refused and the limit is stated.

In every alternate flow the transaction is rolled back. No registration is created and no seat is consumed.

Postconditions	Either the registration exists and the section has one fewer seat, or nothing changed and the student has been told why.
Business Rules	**BR6:** All prerequisites must be completed before registering. **BR7:** Registrations may never exceed a section's capacity. **BR8:** A student may not hold two registrations at overlapping times in the same block. **BR9:** Course load per block is limited by student category.

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5. Non-Functional Requirements

ID	Requirement
NFR1	Schedule and registration pages respond within 2 seconds under normal load.
NFR2	150 concurrent users are supported during a registration window.
NFR3	Section capacity is enforced correctly under concurrent registration. The last seat is granted to exactly one student.
NFR4	Every operation is authorized by role; a student may access only their own registrations.
NFR5	Student pages are usable at 375 px width; all traffic uses TLS.
NFR6	The system runs on any platform with a Java 11+ runtime and a relational database.

6. Traceability: Features to Use Cases

Vision feature	Use cases
F1 Faculty Profile Management	UC1
F2 Course & Program Catalog	UC2
F3 Schedule Generation	UC3
F4 Faculty Assignment & Conflict Detection	UC3
F5 Online Student Registration	UC4, UC5
F6 Registration Rule Enforcement	UC4, UC5
F7 Schedule & Enrolment Views	UC6
F8 Authentication & Role-Based Access	all